

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
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Experiment 3: Regression Analysis using Linear and Regularized Models

1. Aim and Objective

The primary objective of this study is to construct and evaluate multiple regression architectures, specifically linear, Ridge, Lasso, and Elastic Net models, to forecast a continuous target variable. The experiment seeks to analyze predictive accuracy across various metrics and investigate the trade-offs between bias and variance through hyperparameter optimization.

2. Dataset

The investigation utilizes a real-world financial dataset containing diverse numerical and categorical attributes related to loan applications. The goal is to accurately predict the **loan amount sanctioned** for each applicant based on their profile.

3. Preprocessing Steps

A comprehensive data transformation pipeline was established to ensure that the input features were properly formatted for linear modeling.

Handling Missing Values

Numerical Data

Missing entries in numerical columns were addressed using a *median* imputation strategy, which provides a measure of central tendency that is less sensitive to extreme outliers.

Categorical Data

For categorical features, missing observations were replaced using the *most_frequent* value (mode) of the respective column to maintain the categorical distribution.

Encoding Categorical Variables

Categorical variables were transformed into a machine-readable format via **One-Hot Encoding**. The encoder was configured to ignore unknown categories during the testing phase to prevent execution errors.

Feature Scaling

Standardization was performed on all numerical features using **StandardScaler**, centering the data at a zero mean and scaling to unit variance. This ensures that regularization penalties are applied equitably across features with different original scales.

4. Implementation Details

The study compared a baseline linear model against several regularized variants to determine the most effective approach for this dataset.

Regression Models

Linear Regression (Baseline)

The baseline model was established using standard Ordinary Least Squares (OLS) without any regularization penalties.

Ridge Regression (L2 Regularization)

Ridge models were implemented to prevent coefficient explosion by adding a squared magnitude penalty ($L2$). The tuning grid explored $\alpha \in \{0.01, 0.1, 1, 10, 100\}$.

Lasso Regression (L1 Regularization)

Lasso regression was utilized to encourage feature sparsity through an absolute magnitude penalty ($L1$). The grid search evaluated $\alpha \in \{0.001, 0.01, 0.1, 1, 10\}$.

Elastic Net Regression

This hybrid model combined both $L1$ and $L2$ penalties. The hyperparameters tuned included $\alpha \in \{0.01, 0.1, 1, 10\}$ and $l1_ratio \in \{0.2, 0.5, 0.8\}$.

Validation Strategy

K-Fold Cross-Validation

A 5-Fold Cross-Validation approach was used to ensure that the selected hyperparameters provided stable performance across different data subsets.

Scoring Metric

The models were optimized based on the R^2 score (Coefficient of Determination), representing the proportion of variance captured by the model.

5. Visualizations

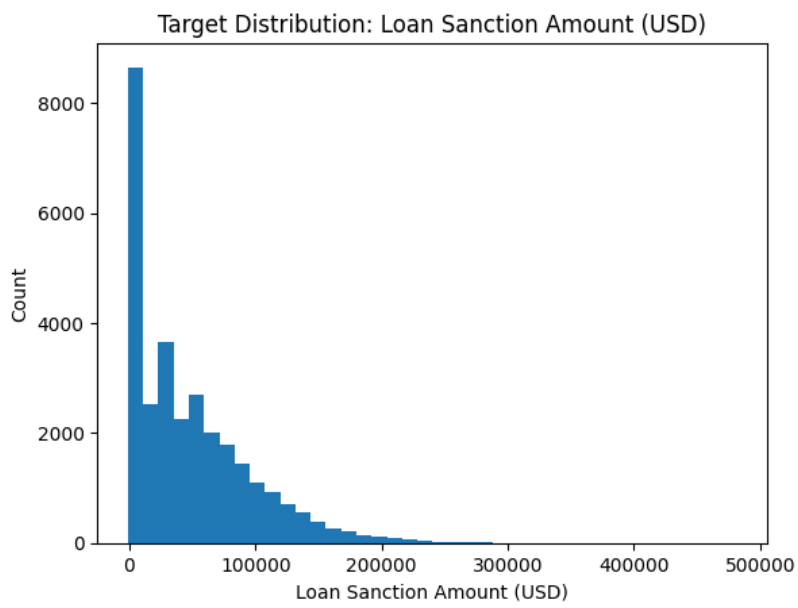


Figure 1: Target Distribution

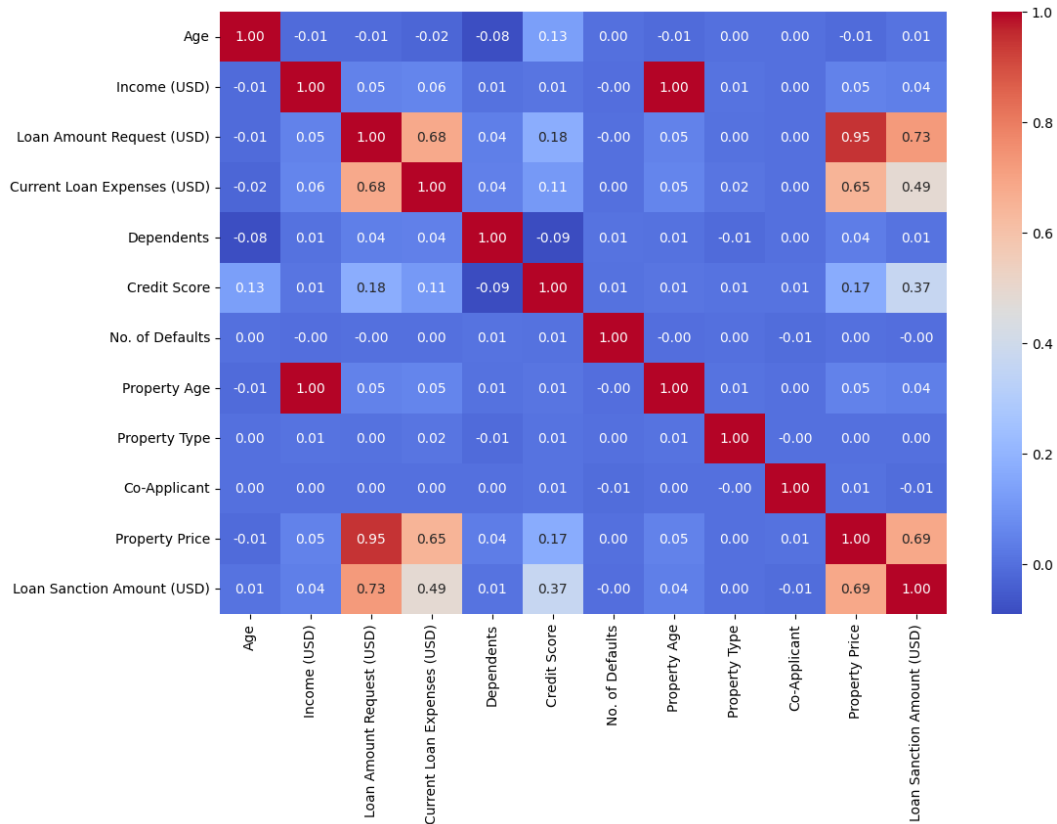


Figure 2: Correlation

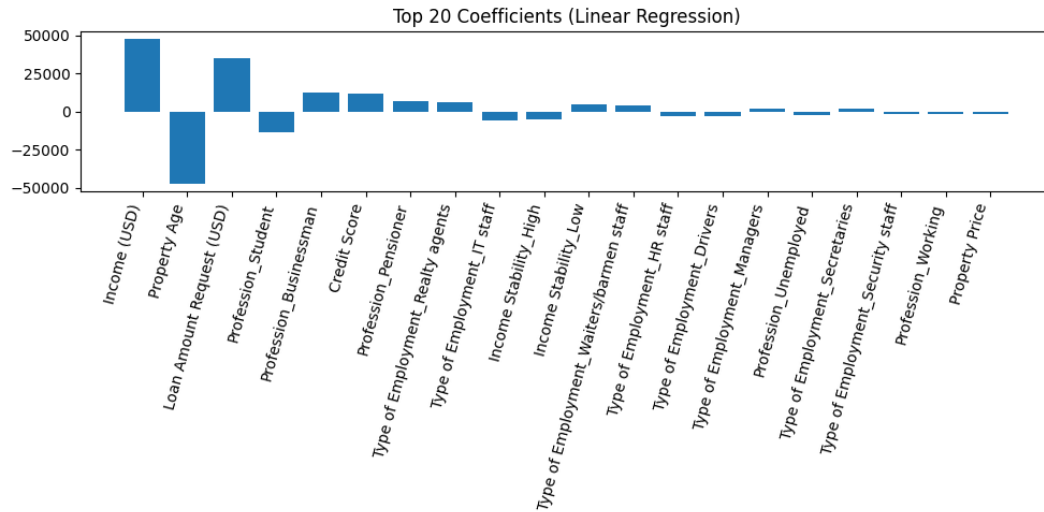


Figure 3: Coefficient comparison bar plot

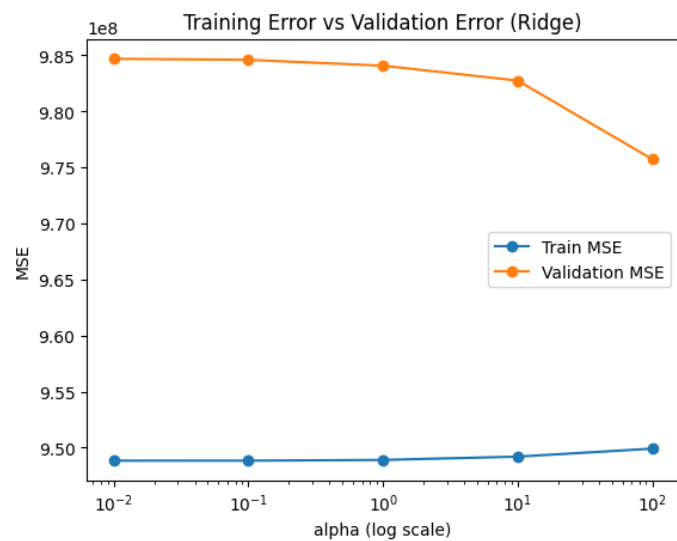


Figure 4: Training error vs. validation error plot

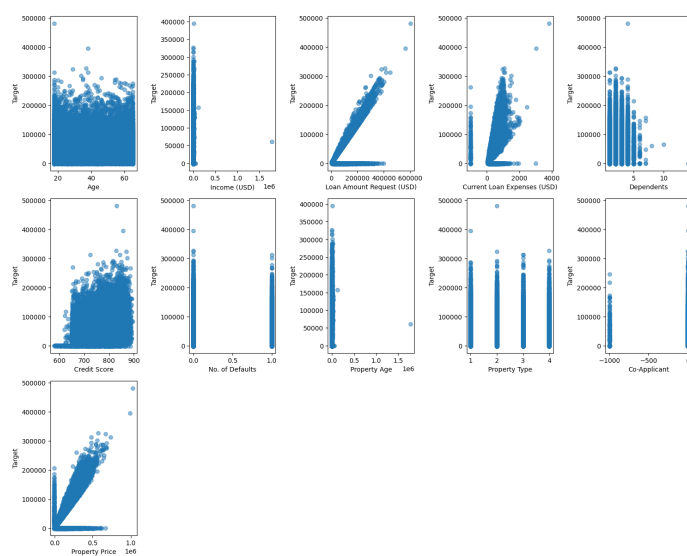


Figure 5: Scatter plot of various features

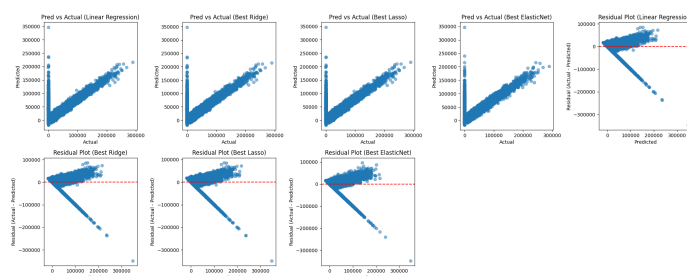


Figure 6: predicted vs actual and residual plots

6. Performance Tables

Hyperparameter Tuning Results

Table 1: Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	Best CV R^2
Ridge Regression	Grid	alpha: 100	0.5786
Lasso Regression	Grid	alpha: 10	0.5785
Elastic Net Regression	Grid	alpha: 0.1, l1_ratio: 0.5	0.5798

Cross-Validation Performance (K = 5)

Table 2: Cross-Validation Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression	21508.1142	984727093.1145	31380.3616	0.5790
Ridge Regression	21524.7473	983019851.6404	31353.1474	0.5797
Lasso Regression	21497.8842	983433813.8426	31359.7483	0.5796
Elastic Net Regression	21732.7267	980735482.7595	31316.6965	0.5807

Test Set Performance Comparison

Table 3: Test Set Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression	21589.8579	1.018912e+09	31920.3984	0.5511
Ridge Regression	21582.3204	1.017227e+09	31893.9988	0.5518
Lasso Regression	21564.5711	1.017745e+09	31902.1118	0.5516
Elastic Net Regression	21751.6900	1.018489e+09	31913.7778	0.5513

Effect of Regularization on Coefficients

Table 4: Coefficient Comparison

Feature	Linear	Ridge	Lasso	Elastic Net
Feature 1	-611.2941	-580.5143	-576.6711	-558.6418
Feature 2	47656.5397	1039.4816	3.8451	97.4573
Feature 3	35365.3700	33825.7866	35169.9575	25155.4719

7. Overfitting and Underfitting Analysis

Training vs. Validation Gap

The cross-validation R^2 scores, peaking at approximately 0.5807 for the Elastic Net model, are consistently higher than the test set R^2 scores, which hover around 0.551. Although there is a visible reduction in performance on unseen data, the generalization gap remains small (approximately 0.03), suggesting that the models are stable and do not suffer from significant overfitting.

Underfitting Analysis

Despite the stability of the models, the test R^2 score of roughly 0.5518 (Ridge) indicates that the models account for only about 55

Effect of Regularization

The variance in test R^2 scores between the baseline Linear Regression (0.5511) and the best-performing regularized model, Ridge Regression (0.5518), is statistically negligible. This confirms that high variance was not the primary hindrance to performance; rather, the models were limited by their representational capacity (bias).

8. Bias–Variance Analysis

Bias Behavior

The high bias inherent in linear models is evident here, as they fail to capture the full complexity of the financial dataset. This results in a relatively high Mean Absolute Error (MAE) of approximately 21,589 for the baseline model, indicating consistent, systematic prediction errors.

Variance Reduction

- **Linear Regression:** The baseline model demonstrated high variance in its coefficient estimates. For instance, **Feature 2** was assigned an excessively large coefficient of 47,656.5397, making the model highly sensitive to minor input fluctuations.
- **Ridge and Elastic Net Regression:** Regularization effectively mitigated this instability. Ridge Regression significantly constrained the **Feature 2** coefficient to 1,039.4816, while Elastic Net brought it to 97.4573. This shrinkage improves model robustness even when accuracy gains are modest.

Feature Sparsity (Lasso Regression)

Lasso Regression successfully introduced sparsity by driving the coefficient of **Feature 2** down to a mere 3.8451, effectively minimizing the influence of less informative variables. This behavior simplifies the model and highlights which applicant attributes are truly essential for predicting the sanctioned loan amount.

9. Observations and Conclusion

Feature Importance

The analysis indicates that *Loan Amount Request (USD)* and *Credit Score* serve as the primary predictors in this dataset. Correlation metrics further support this, showing that the requested amount maintains the strongest positive relationship with the final sanctioned amount.

Model Performance

All four regression architectures performed similarly, with Ridge Regression providing the marginal lead on the test set with an R^2 of 0.5518. The consistent performance across all models suggests that the predictive power of linear relationships for this specific dataset has reached its ceiling.

Conclusion

While regularization proved highly effective at stabilizing model parameters and preventing coefficient "explosion," it could not overcome the fundamental underfitting caused by the linear model constraint. To achieve higher predictive accuracy (surpassing the 0.60 R^2 threshold), future iterations should consider non-linear approaches such as Random Forests, Support Vector Regression (SVR), or Gradient Boosting.

Dataset reference

- Kaggle: Predict Loan Amount Data

References

- Scikit-learn: Linear Models
- Scikit-learn: Hyperparameter Optimization
- Loan Amount Dataset